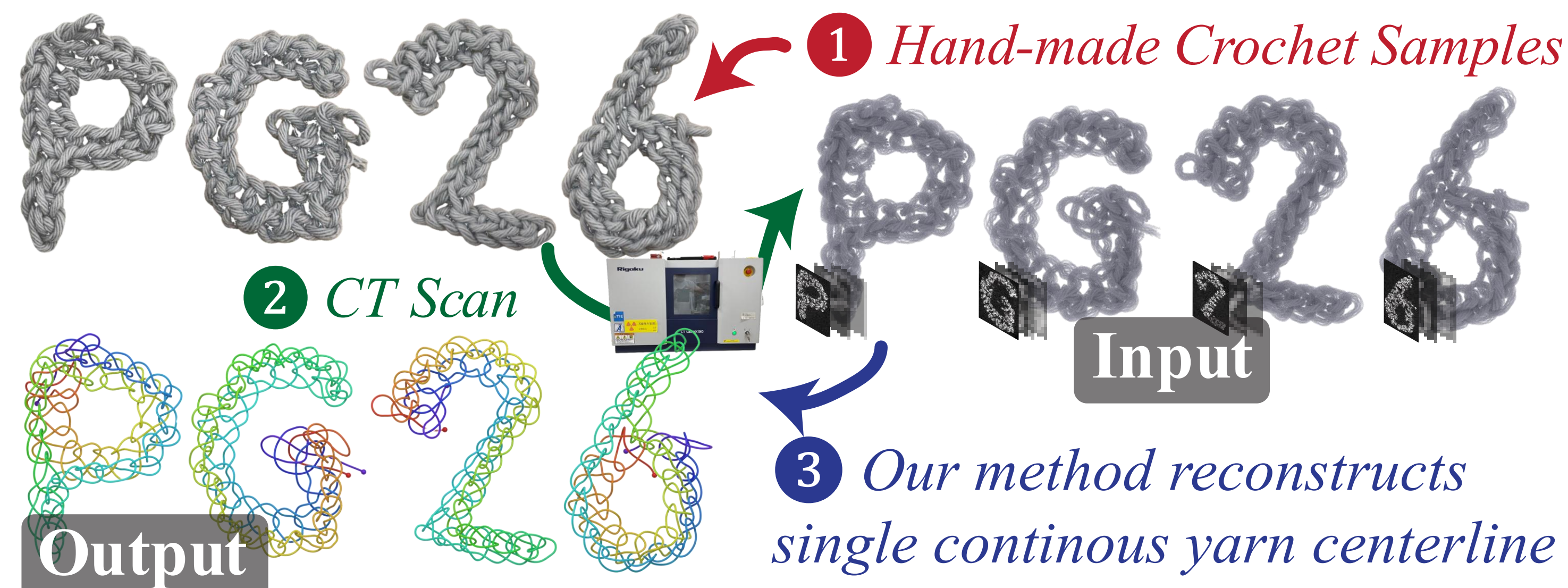


CT2Yarn: Yarn Level Reconstruction of Crochet from Computed Tomography

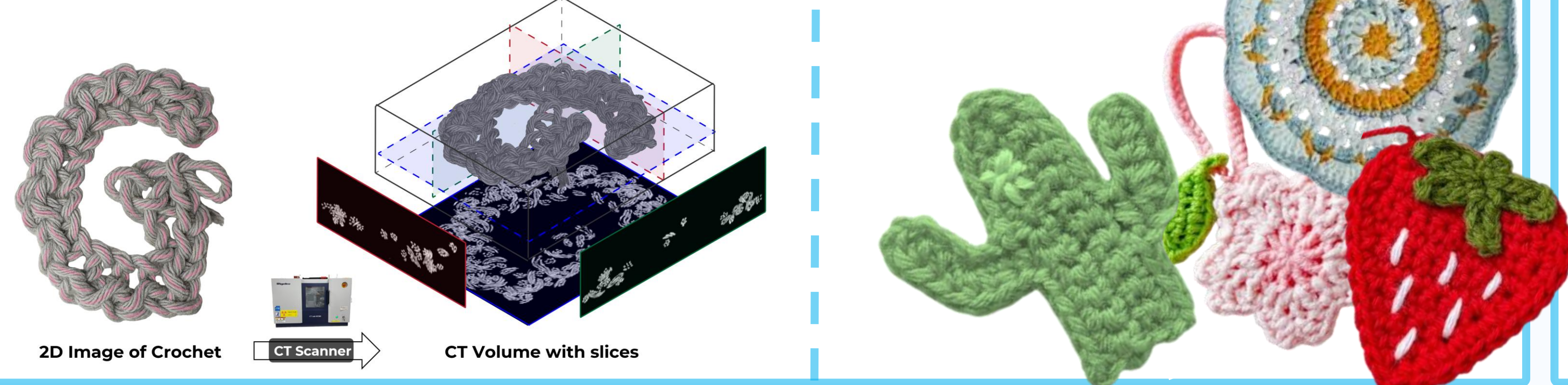
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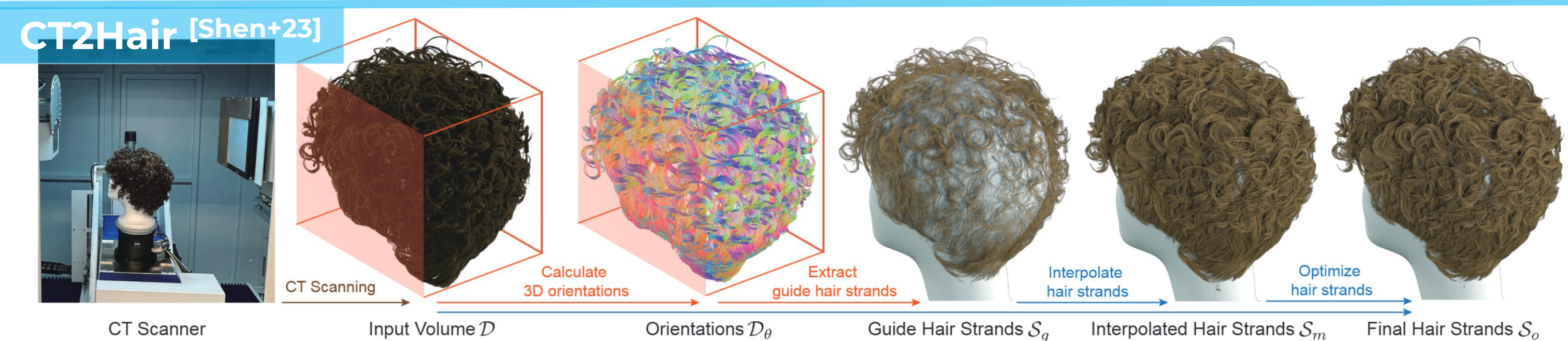
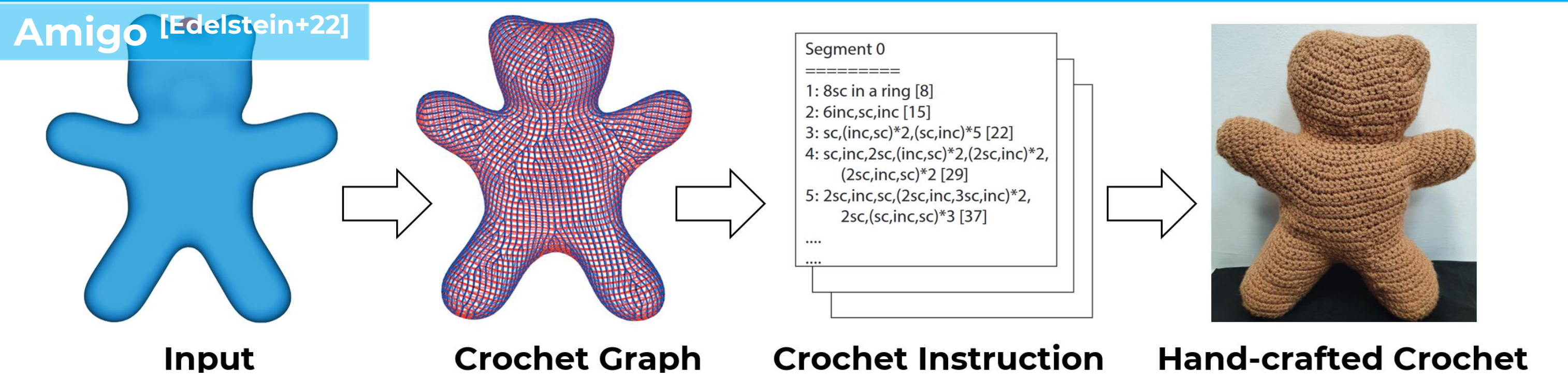
1. Introduction



- Interlocking loops obscure the yarn path.
- Micro-CT reveals the hidden internal structure.

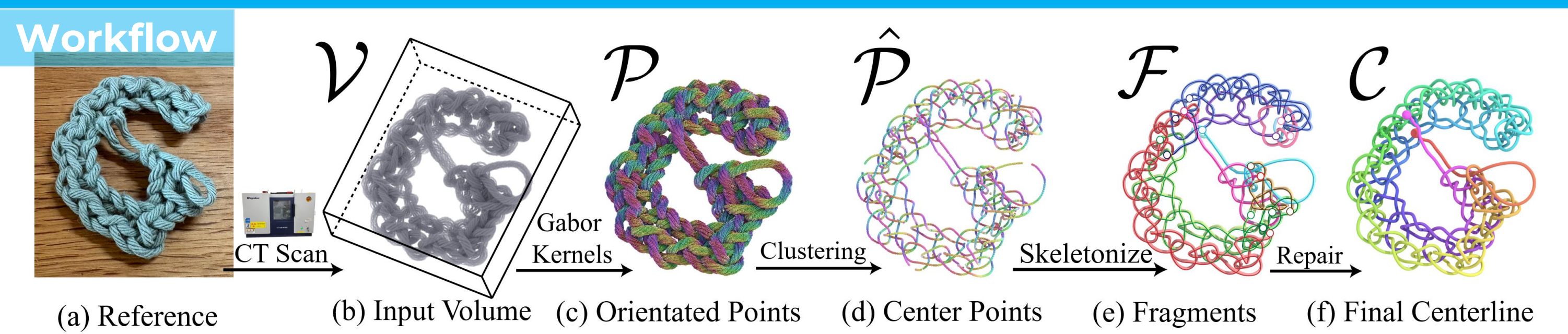


2. Related works

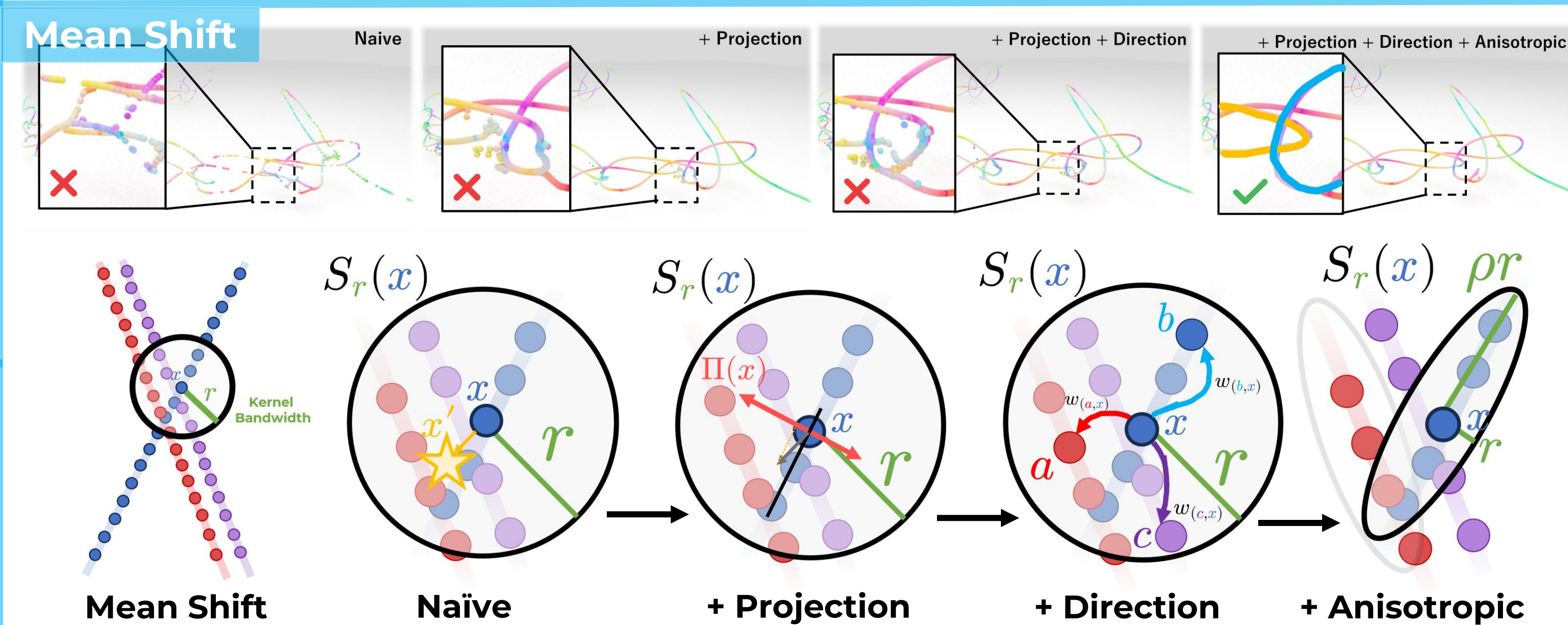
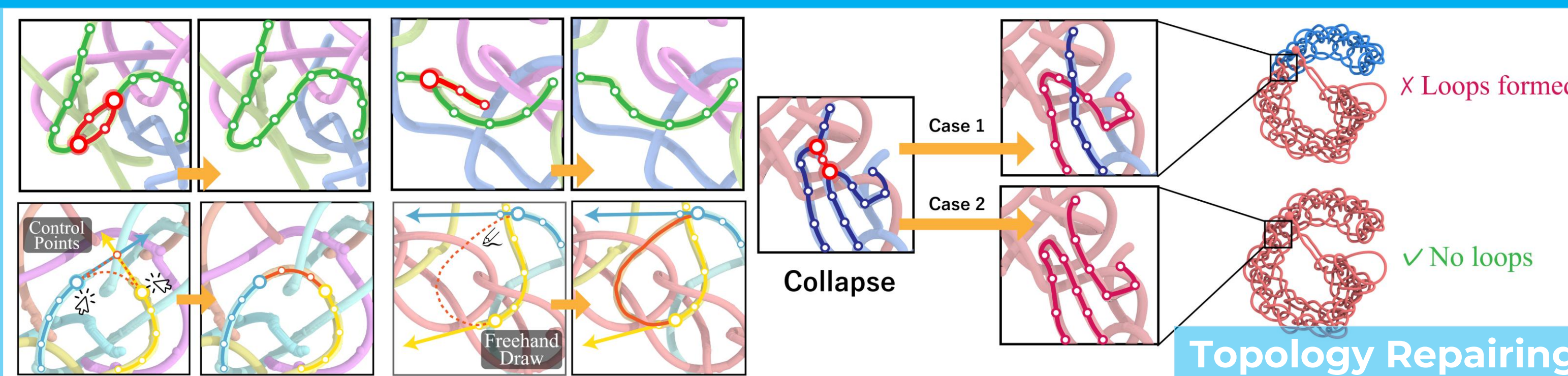
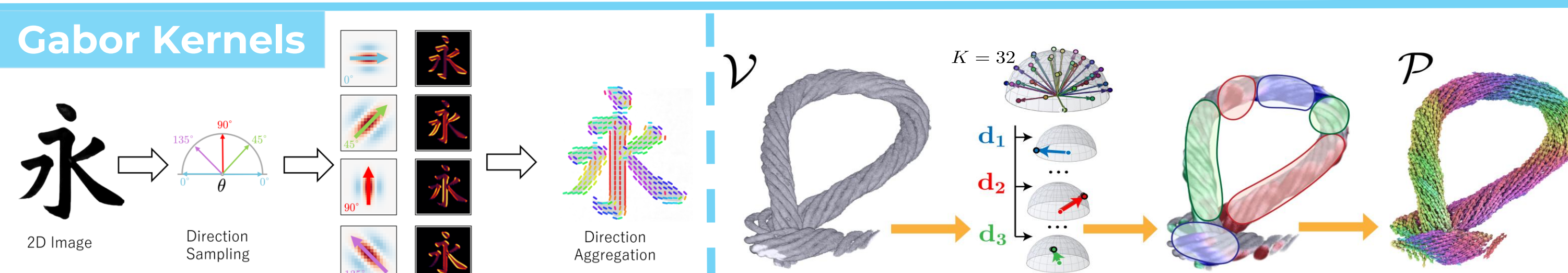


- AmiGo generates crochet patterns;
- CT2Hair reconstructs independent strands.
- Crochet has Hierarchy!
- Local error will lead to error in global topology!

3. Methodology

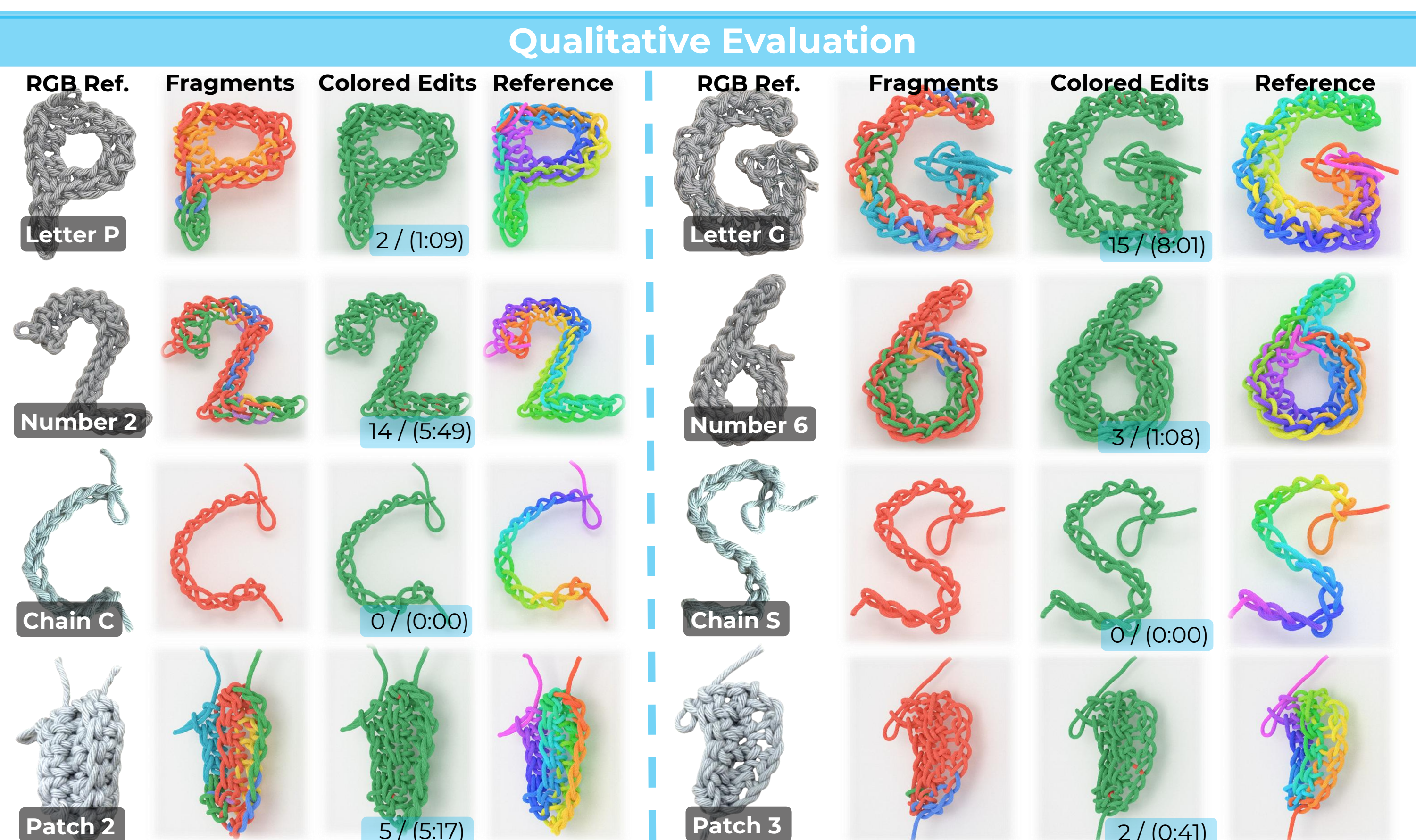


- Orientation estimation** Gabor extracts local fiber direction.
- Center-point extraction**. Anisotropic mean-shift aggregates directions at the yarn scale. Perpendicular projection, direction-aware weighting, and ellipsoid kernels concentrate points toward the yarn centerline.
- Topology reconstruction**. MST connects fragments. Topology repair resolve common breaks with UI repair.

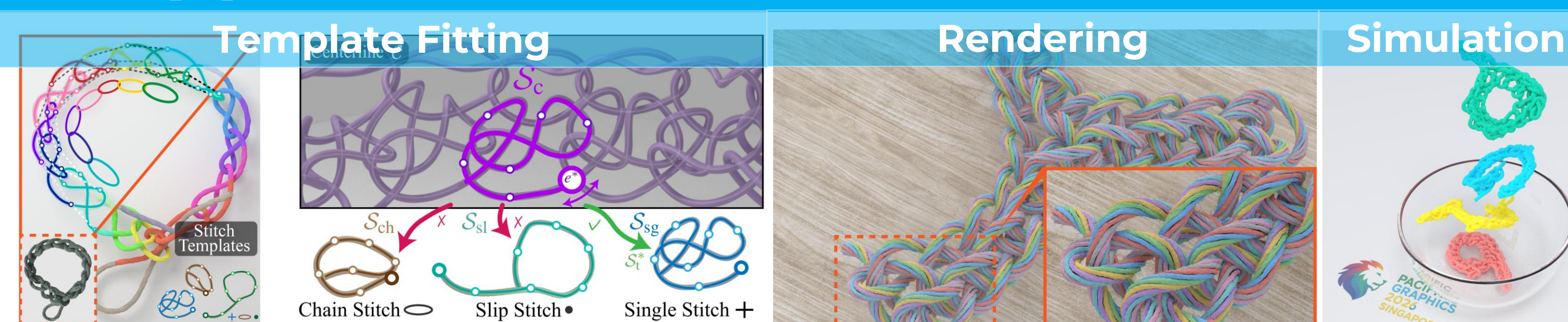


4. Results

Quantitative Evaluation		Reference-free		Reference-based		Effort	
Method	#C	$\bar{L}_C$	$E_L \downarrow$	$\bar{\alpha} \downarrow$	$\ell_{CD} \downarrow$	$\ell_2 \downarrow$	edits time
CT2Hair (Gab.)	88	314	0.115	10.55	16.99	18.55	— —
CT2Hair (Grad.)	115	236	0.271	12.75	20.60	24.67	— —
Ours (Auto)	15	3814	0.002	2.66	2.43	2.26	— —
Ours (Manual)	1	30359	—	—	—	—	4 2:46



5. Applications and Limitations



- Template fitting** can extract the semantic stitch pattern.
- Ply geometry with centerline enables realistic rendering.
- Discrete elastic rods support simulation under gravity.

- Tightly crocheted regions** produce more fragmented centerlines and require additional manual repair.

This work was supported by JST SPRING, Grant Number JPMJSP2108.

References:

- Shen, Y., Saito, S., Wang, Z., Maury, O., Wu, C., Hodgins, J., ... & Nam, G. (2023). Ct2hair: High-fidelity 3d hair modeling using computed tomography. *ACM Transactions on Graphics (TOG)*, 42(4), 1-13.
- Edelstein, M., Peleg, H., Itzhaky, S., & Ben-Chen, M. (2022, October). Amigo: Computational design of amigurumi crochet patterns. In *Proceedings of the 7th Annual ACM Symposium on Computational Fabrication* (pp. 1-11).



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